

Objective

Deploy a static website from a GitHub repository to an AWS EC2 Linux VM using a Jenkins Freestyle project. Jenkins will check out the GitHub repository, install/start Apache HTTP Server if required, copy the website files to /var/www/html, and verify that the website is accessible through the EC2 public IP.

Architecture / Flow

GitHub Repository → Jenkins Freestyle Job → Git Checkout → Jenkins Workspace → Install/Start Apache → Deploy Files → /var/www/html → Apache :80 → Browser

Item	Value
Source	GitHub repository
Repository	https://github.com/Shantanumajan6/vel-app.git
Branch	master
Jenkins Job	GitHub-Apache-Deployment
Jenkins Type	Freestyle Project
Destination	AWS EC2 Linux VM
Web Server	Apache HTTP Server (httpd)
Document Root	/var/www/html
HTTP Port	80

Jenkins Freestyle Deployment Steps

1. Verify the Linux VM

Connect to the EC2 VM using SSH.

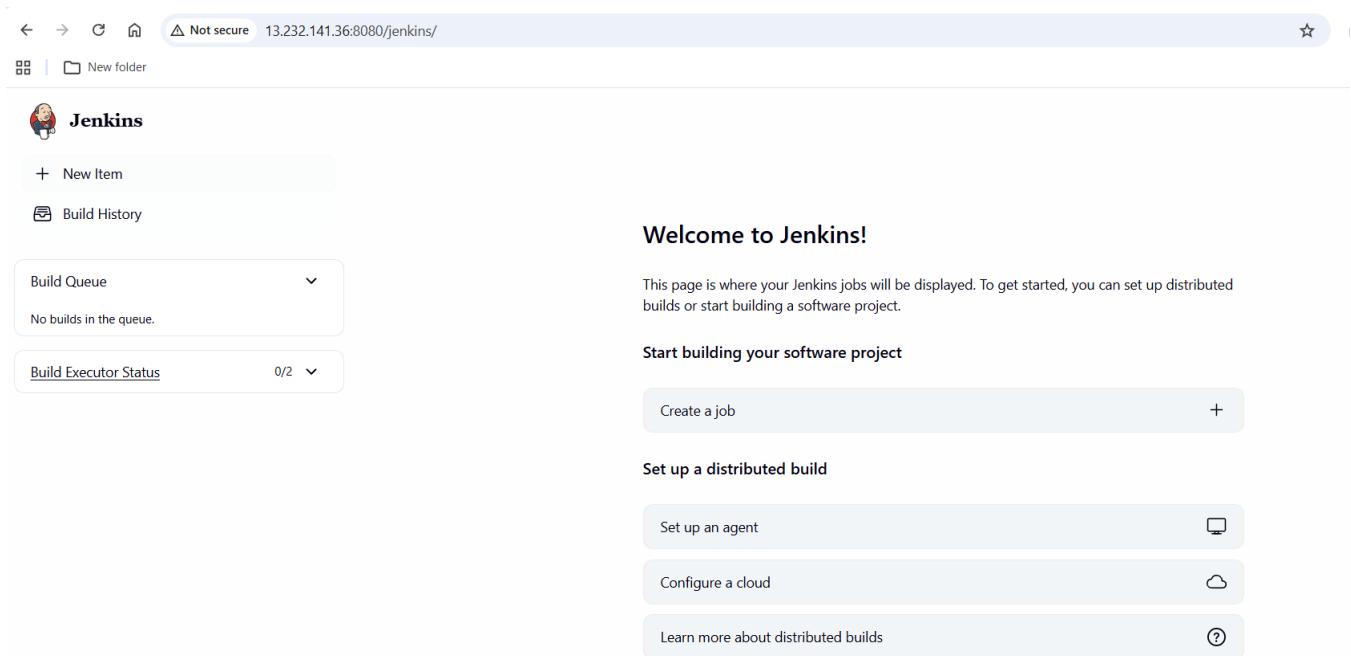
```
ssh -i "key_pair.pem" ec2-user@ec2-13-232-141-36.ap-south-1.compute.amazonaws.com
```

Note: Apache installation and service management will be performed by the Jenkins Freestyle build.

2. Install/Verify Jenkins

If Jenkins is already installed on your VM, verify it is running. Since you are using Jenkins Freestyle, create a Freestyle project.

```
[ec2-user@ip-172-31-6-238 bin]$ cd /mnt/servers/apache-tomcat-10.1.60/bin
./startup.sh
Using CATALINA_BASE:   /mnt/servers/apache-tomcat-10.1.60
Using CATALINA_HOME:   /mnt/servers/apache-tomcat-10.1.60
Using CATALINA_TMPDIR: /mnt/servers/apache-tomcat-10.1.60/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /mnt/servers/apache-tomcat-10.1.60/bin/bootstrap.jar:/mnt/servers/apache-tomcat-10.1.60/bin/tomcat-juli.jar
Tomcat started.
[ec2-user@ip-172-31-6-238 bin]$
```



3. Install and Verify Git

```
yum install git -y
git --version
```

Git must be available on the Jenkins VM so that Jenkins can check out the GitHub repository.

4. Configure GitHub Repository in Jenkins

The GitHub repository will be checked out automatically by Jenkins.

Repository: <https://github.com/Shantanumajan6/vel-app.git>

Branch: */master

Note: Manual git clone is not required. Jenkins performs the Git checkout before executing the build steps.

5. Create Jenkins Freestyle Project

In Jenkins:

1. Click New Item
2. Enter: GitHub-Apache-Deployment
3. Select Freestyle project
4. Click OK

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🗃 New folder

 **Jenkins** / New Item

New Item

Enter an item name

GitHub-Apache-Deployment

Select an item type



Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK

6. Configure Source Code Management

Select: Source Code Management → Git

Repository URL: <https://github.com/Shantanumajan6/vel-app.git>

Branch: */master

The repository contains: index.html and dev.html

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🗂️ New folder

Jenkins / GitHub-Apache-Deploym... / Configure

Configure

- ⚙️ General
- 🔑 Source Code Management**
- 🕒 Triggers
- 🌐 Environment
- 📋 Build Steps
- 🔗 Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ **Git** ?

Repositories ?

Repository URL ?

Credentials ?

- none -

+ Add

Advanced ▾

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

7. Add Jenkins Execute Shell Build Step

Go to: Build → Add build step → Execute shell

Use:

```
#!/bin/bash

echo "==== Starting Deployment ====="

echo "==== Jenkins Workspace ====="
pwd
ls -la

echo "==== Installing Apache ====="
sudo yum install httpd -y

echo "==== Starting Apache ====="
sudo systemctl enable httpd
sudo systemctl start httpd

echo "==== Deploying Website ====="
sudo cp -r ./ * /var/www/html/

echo "==== Setting Permissions ====="
sudo chown -R apache:apache /var/www/html
sudo chmod -R 755 /var/www/html

echo "==== Deployed Files ====="
ls -la /var/www/html/

echo "==== Apache Status ====="
sudo systemctl status httpd --no-pager

echo "==== Testing Website ====="
curl -I http://localhost
```

```
echo "==== Deployment Completed ====="
```

Jenkins

GitHub-Apache-Deploym...

Status

Changes

Console Output

Delete build #1

Git Build Data

Edit build information

Download

Copy

View as plain text

Started by user Admin

Running as SYSTEM

Building in workspace /root/.jenkins/workspace/GitHub-Apache-Deployment

The recommended git tool is: NONE

No credentials specified

Cloning the remote Git repository

Cloning repository https://github.com/ShantanuJank/vel-app.git

> git init /root/.jenkins/workspace/GitHub-Apache-Deployment # timeout=10

Fetching upstream changes from https://github.com/ShantanuJank/vel-app.git

> git --version # timeout=10

> git --version # 'git version 2.50.1'

> git fetch --tags --force --progress -- https://github.com/ShantanuJank/vel-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git config remote.origin.url https://github.com/ShantanuJank/vel-app.git # timeout=10

> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10

Avoid second fetch

> git rebase refs/remotes/origin/master^{commit} # timeout=10

Checking out Revision f93e73aca2a383cf632908b073e12b13ac80cf8 (refs/remotes/origin/master)

> git config core.sparsecheckout # timeout=10

> git checkout -f f93e73aca2a383cf632908b073e12b13ac80cf8 # timeout=10

Commit message: "Revert "added c3""

First time build. Skipping changelog.

([GitHub-Apache-Deployment]) \$ /bin/bash /mnt/servers/apache-tomcat-10.1.60/temp/jenkins812471657777678510.sh

==== Starting Deployment =====

==== Jenkins Workspace =====

/root/.jenkins/workspace/GitHub-Apache-Deployment

total 8

drwxr-xr-x. 3 root root 52 Sep 28 16:58 .

drwxr-xr-x. 3 root root 38 Sep 28 16:58 ..

drwxr-xr-x. 7 root root 146 Sep 28 16:58 .git

-rwxr-xr-x. 1 root root 10 Sep 28 16:58 dev.html

-rwxr-xr-x. 1 root root 12 Sep 28 16:58 index.html

==== Installing Apache =====

Last metadata expiration check: 1:57:21 ago on Mon Sep 28 15:01:32 2026.

Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.

Dependencies resolved.

Nothing to do.

Complete!

==== Starting Apache =====

Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.

==== Deploying Website =====

==== Setting Permissions =====

==== Deployed Files =====

total 8

drwxr-xr-x. 2 apache apache 40 Sep 28 16:58 .

drwxr-xr-x. 4 root root 33 Sep 28 15:01 ..

-rwxr-xr-x. 1 apache apache 10 Sep 28 16:58 dev.html

-rwxr-xr-x. 1 apache apache 12 Sep 28 16:58 index.html

==== Apache Status =====

● httpd.service - The Apache HTTP Server

Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)

Active: active (running) since Mon 2026-09-28 16:58:54 UTC; 214ms ago

Docs: man:httpd.service(8)

Main PID: 32091 (httpd)

Status: "Started, listening on: port 80"

Tasks: 177 (limit: 2899)

Memory: 19.1M

CPU: 74ms

CGroup: /system.slice/httpd.service

└─32091 /usr/sbin/httpd -DFOREGROUND

└─32092 /usr/sbin/httpd -DFOREGROUND

└─32093 /usr/sbin/httpd -DFOREGROUND

└─32098 /usr/sbin/httpd -DFOREGROUND

└─32099 /usr/sbin/httpd -DFOREGROUND

Set 28 16:58:54 ip-172-31-6-238.ap-south-1.compute.internal system[1]: Starting httpd.service - The Apache HTTP Server...

Set 28 16:58:54 ip-172-31-6-238.ap-south-1.compute.internal system[1]: Started httpd.service - The Apache HTTP Server.

Set 28 16:58:54 ip-172-31-6-238.ap-south-1.compute.internal http[32091]: Server configured, listening on: port 80

==== Testing Website =====

% Total % Received % Xferd Average Speed Time Time Time Current

Download Upload Total Spent Left Speed

0 0 0 0 0 0 0 0 0 0

0 12 0 0 0 0 0 0 0 0

0 12 0 0 0 0 0 0 0 0

0 12 0 0 0 0 0 0 0 0

HTTP/1.1 200 OK

Date: Mon, 28 Sep 2026 16:58:55 GMT

Server: Apache/2.4.68 (Amazon Linux)

Last-Modified: Mon, 28 Sep 2026 16:58:55 GMT

ETag: W/"c-65c0f365171"

Accept-Ranges: bytes

Content-Length: 12

Content-Type: text/html; charset=UTF-8

==== Deployment Completed =====

Finished: SUCCESS

Jenkins 2.568.3

Jenkins 2.568.3

One Jenkins build now demonstrates the full flow: Git checkout → Apache installation → Apache startup → website deployment → permissions → verification.

8. Verify Deployment on EC2 VM

After a successful Jenkins build, SSH into the VM and run:

```
ls -la /var/www/html/
```

Expected files: index.html, dev.html

Then:

```
curl -I http://localhost
```

Expected: HTTP/1.1 200 OK

```
[root@ip-172-31-6-238 bin]# ls -la /var/www/html/
total 8
drwxr-xr-x. 2 apache apache 40 Sep 28 16:58 .
drwxr-xr-x. 4 root root 33 Sep 28 15:01 ..
-rwxr-xr-x. 1 apache apache 10 Sep 28 17:11 dev.html
-rwxr-xr-x. 1 apache apache 12 Sep 28 17:11 index.html
[root@ip-172-31-6-238 bin]#
```

9. Verify Website from Browser

Open: <http://13.232.141.36/index.html>

You can also test: <http://13.232.141.36/index.html>

